

EDUCATION

Guru Nanak Dev University, Amritsar	(July 2024 - July 2028)
B.Tech CSE - 8.0/10	
Spring Dale Senior School, Amritsar	(March 2017 - March 2024)
10th CBSE - 95% 12th CBSE - 87.5%	

PROJECTS

AI-Driven Medical Report Analysis & Appointment System - [Github](#)

- Engineered an AI-driven medical report analysis pipeline using OCR, NLP, and GROK AI to extract symptoms and generate structured summaries with high accuracy
- Designed a doctor recommendation engine leveraging contextual symptom matching and weighted scoring logic to route patients to relevant specialists
- Built secure RESTful APIs using Django Ninja with JWT-based RBAC (Admin/Doctor/Patient) to support appointment scheduling, notifications, and status workflows
- Optimized backend performance and reliability by integrating Redis caching and Celery-based asynchronous workers, significantly reducing request latency and blocking operations

Tech Stack: Django, Django-Ninja, GROK API, React, Langchain, PostgreSQL, Redis, Celery, OCR

Automated YouTube Shorts Generation Pipeline - [Github](#)

- Engineered an automated video processing pipeline that converts YouTube videos into vertical Shorts by detecting the most engaging segments using OpenCV-based scene and activity detection
- Built an AI-powered subtitle generation system using OpenAI Whisper with ASS-based advanced styling, enabling clean, professional captions with customizable fonts, colors, and sizes
- Developed optimized backend workflows for video downloading, trimming, resizing, background blur, and filtering using FFmpeg, ensuring efficient and high-quality media processing
- Designed a responsive React frontend with real-time subtitle preview and styling controls, delivering an intuitive and interactive user experience

Tech Stack: Django, Django-Ninja, React, JavaScript, TailwindCSS, PostgreSQL, OpenCV, Ffmpeg, Whisper

Location-Based Farmers E-Commerce Platform - [Github](#)

- Built a location-based e-commerce platform connecting users with nearby farmers to enable access to fresh produce at lower costs
- Developed RESTful backend services using Django Ninja to support product management, user workflows, and scalable API interactions
- Integrated Google Maps API to allow users to discover nearby sellers based on real-time location data
- Designed a robust data layer using PostgreSQL to manage products, users, and transactions with consistency and reliability

Tech Stack: Django, Django-Ninja, JavaScript, Bootstrap, PostgreSQL, Google-Maps API , render

SKILLS

Languages: Python, JavaScript, C++, HTML/CSS
Frameworks: Django, Django-Ninja, React, Next Js, Langchain
Technologies: REST APIs, OpenAI API, Celery, JWT, Github, Docker, Google ADK
Databases: PostgreSQL, SQLite , Redis, Supabase

ACHIEVEMENTS

- 2nd Position – HACKERWRATH 2.0:** Built an AI + Human-supervised chatbot system using RAG, custom iframes, real-time chat, a human oversight dashboard, and an integrated sentiment analysis model.
- 1st Position – Node Hash:** Built an online multimedia platform for queuing daily tasks, tracking progress, and generating monthly reports.